Education Management System

Sprint 1 Project Foundation - Team 23

Team Members and Roles (RACI Matrix)

Full Name	Student ID	University Email	Role	Sprint 1 Responsibilities
Alperen Gökay Göktaş (Killly)	220201040	s220201040@ankarabilim.edu.tr	Product Owner	Project vision, Backlog (A), Acceptance criteria (A)
Yağmur Deniz Biçer	230201051	s230201051@ankarabilim.edu.tr	Scrum Master	Process management, Repo rules (A)
Berkay Kervan	230204007	s230204007@ankarabilim.edu.tr	Lead Software Architect	Architecture design, OpenAPI spec (A)
Serhat Eroğlu	260201311	s260201311@ankarabilim.edu.tr	QA & Test Automation	Sprint 1 evidence verification (A)
Reyhan Çetin	230201056	s230201056@ankarabilim.edu.tr	DevOps & CI/CD Engineer	DevOps infrastructure (R)
Muhammed Ökmen	260204316	s260204316@ankarabilim.edu.tr	Implementation Engineer	Repo setup, OpenAPI writing (R)

(R: Responsible, A: Accountable, C: Consulted, I: Informed)

1. Project Proposal

Problem Definition

In educational institutions, class capacities, teacher schedules, and student pass/fail statuses based on various exam types (midterm/final/makeup) are often managed through disconnected Excel files. This leads to capacity double-booking conflicts and data integrity errors during end-of-term grading and report card generation.

Proposed Outcome

An API-first education management system that allows administrators to assign classes, enables teachers to enter attendance and grades for their specific classes, and automatically calculates report cards based on configured exam weights (e.g., 40% midterm, 60% final).

First Release Scope

- **Included:** Branch/class capacity management, teacher assignments, student registrations, attendance tracking, flexible exam type configuration, grade weighting, and digital report card generation.
- **Deferred (Out-of-Scope):** Accounting and installment tracking, SMS/WhatsApp parent notifications, and online video class infrastructure.

Assumptions and Risks

- **Assumption:** The system will handle secure authentication locally using username/password within its own database, without needing an external Identity Provider (SSO).
- **Risk:** Concurrent registration requests may lead to capacity conflicts. This will be isolated and managed via integration testing (QA).

Open Source License

MIT License: Chosen because it is a flexible model that permits commercial use, modification, and distribution.

2. System Vision and Boundary

System Vision: The Education Management System automates assignment processes to provide administrative staff with reliable reservation rules; it allows teachers to monitor grade data and enables students to view their weighted report cards.

System Boundary

- **Inside the Software Responsibility:** Capacity catalog, student assignment lifecycle, local authentication, exam configurations, pass/fail algorithms, and report card generation.
- **Outside or Deferred:** Physical classroom maintenance, financial and billing operations.

System Context Interactions

- **Manager (Admin):** Creates classes, defines capacities and exam weights.
- **Teacher:** Enters grades and attendance only for assigned classes.
- **Student/System:** Authenticates and retrieves weighted averages.

3. GitHub Repository Workflow (Git-Flow)

Repository URL: https://github.com/K1llly/SENG-321-PROJECT

- `main`: Production-ready release tags.
- `develop`: Stable development code integrated during the sprint.
- `feature/issue-name`: Short-lived branches based on specific Issues.

Rules: Direct pushes to main branches are prohibited. Every change must be linked to an Issue, and Pull Requests (PRs) must be approved by at least one team member (Peer review) before merging.

4. Initial Product Backlog

ID	User Story	Acceptance Criteria	Effort
US-01	As a user, I want to log in with my username and password to access the system.	Successful login generates a role-based token; invalid credentials return a 401 error.	3
US-02	As a manager, I want to create a class and define its maximum capacity.	Capacity cannot be negative and must be saved to the DB.	5
US-03	As a teacher, I want to enter exam types and grades for students in my assigned classes.	Grades must be between 0-100; unauthorized access must be blocked.	8
US-04	As the system, I want to calculate the weighted grade average and determine the pass/fail status.	Weights (midterm/final) are summed up; below-threshold scores yield a "fail" status.	8